Integrated Communication Circuits

Lecture 4 : Frequency Synthesizer

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第四章 频率合成器

- 尽管晶体振荡器能提供高稳定度的振荡频率，但其频率值是单一的，最多只能在很小的频率段内进行调节。
- 在现代无线通信系统中，接收和发射信号是多信道的。频率合成器可以产生与基准参考频率源具有同样高精度和稳定度的一系列频率信号，它作为接收和发射信号的本振源是通信电路中一个极其重要的组成部件。

§4.1 直接频率合成

频率合成器的主要技术指标

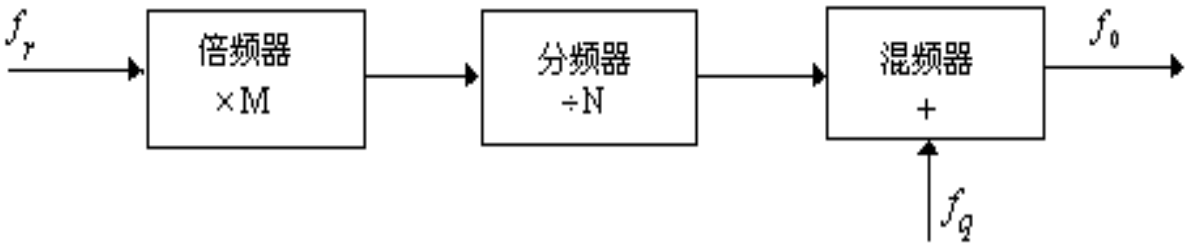
1. 工作频率范围：频率合成器输出的最低频率$f_{\min}$至最高频率$f_{\max}$的范围。

称 $$\boxed{k = \frac{f_{\max}}{f_{\min}}}$$ 为频率覆盖系数。

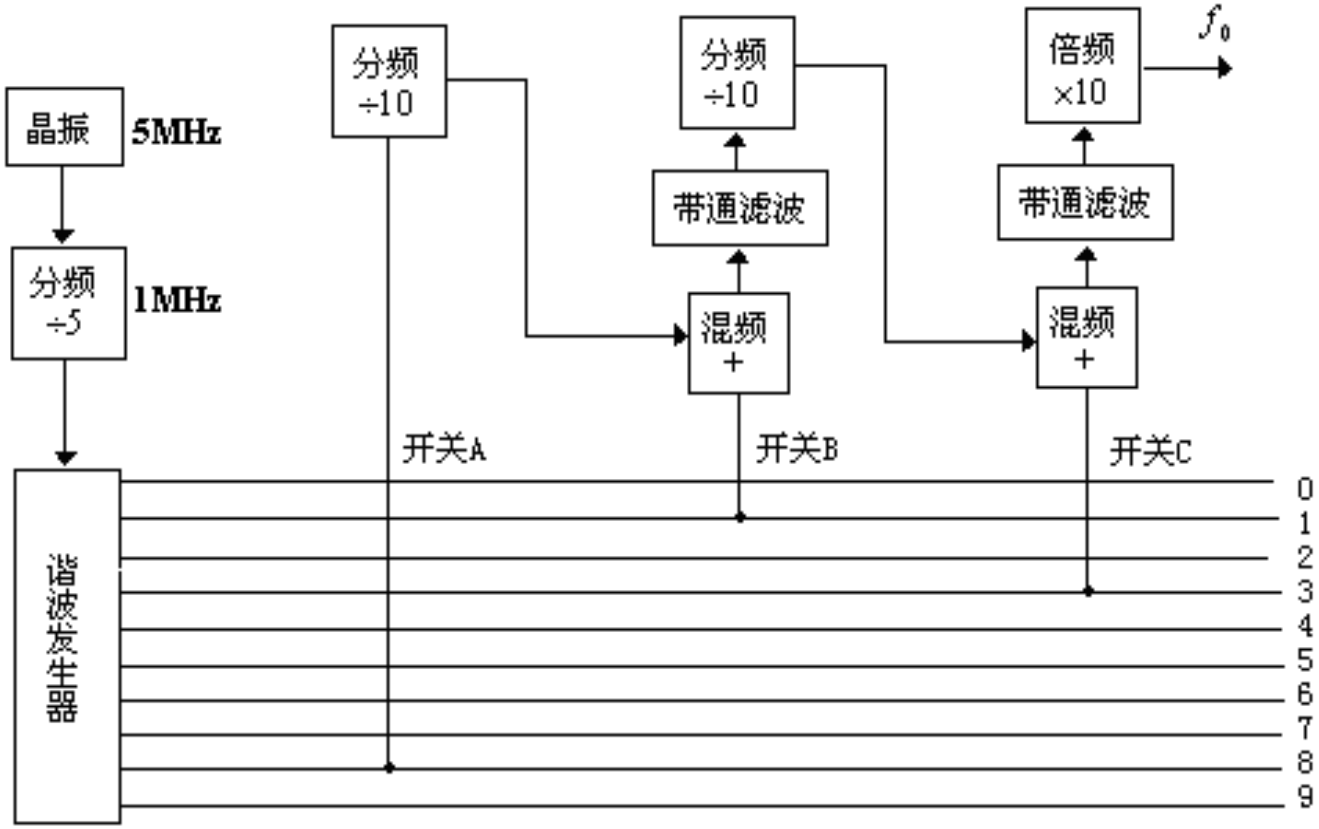
2. 频率间隔 f_{ch}：频率合成器的输出频谱是离散的，两个相邻频率之间的最小间隔称为频率间隔，也可以用 Δf 表示。

直接频率合成法

简单的模拟直接频率合成器组成方框图



$$f_0 = f_q + \frac{M}{N} f_r$$



一个实际的频率合成器的方框图

§4.2 锁相环

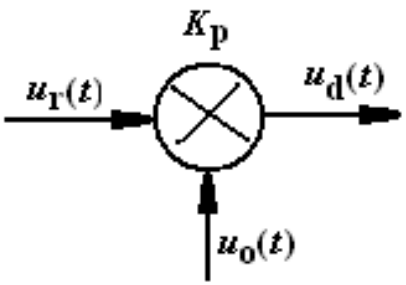
PLL (Phase Lock Loop)

锁相环(PLL)是一个相位误差控制系统。是将参考信号与输出信号之间的相位进行信号比较。产生相位误差电压来调整输出信号的相位，以达到与参考信号同频的目的。

- 鉴相器(Phase Discriminator，简称PD)
- 低通滤波器(Lowpass Filter，或环路滤波, 简称LPF，或LF)
- 压控振荡器(Voltage Control Oscillator，简称VCO)

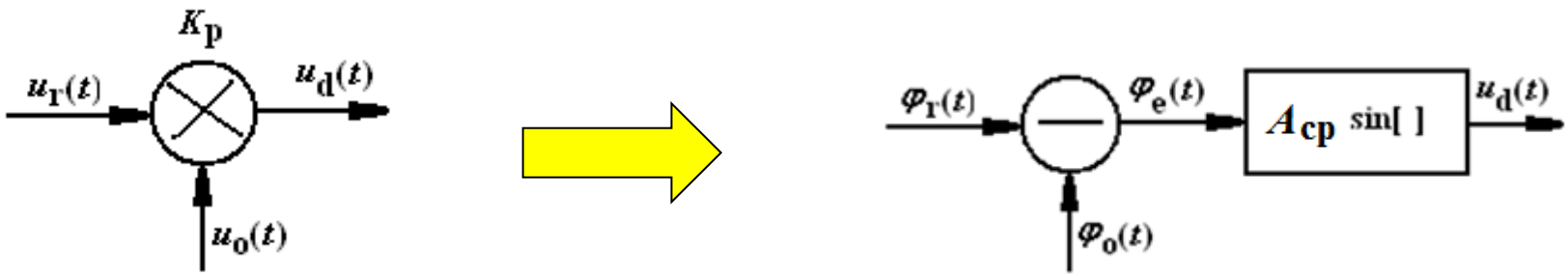
1. 模拟鉴相器

$$u_r(t) = U_{rm}\sin[\omega_r t + \varphi_r(t)]$$



$$u_0(t) = U_{0m}\cos[\omega_0 t + \varphi_0(t)]$$

$$u_r(t)u_0(t) = \frac{K_p}{2}U_{rm}U_{0m}\{\sin[2\omega_0 t + \varphi_1(t) + \varphi_2(t)] + \sin[\varphi_1(t) - \varphi_2(t)]\}$$



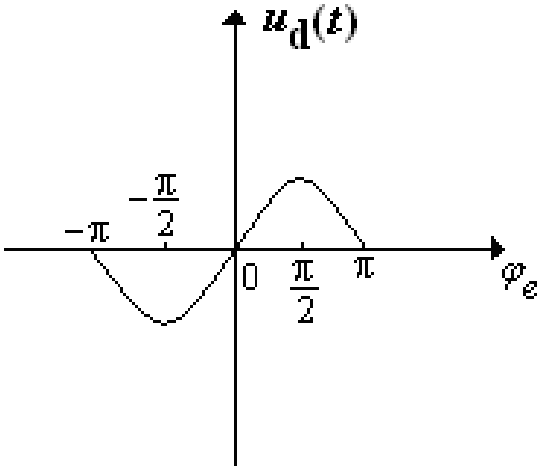
经低通滤波器后，只保留上式第二项：

$$u_d(t) = A_p \sin(\varphi_1(t) - \varphi_2(t)) = A_p \sin(\varphi_e(t))$$

其中：

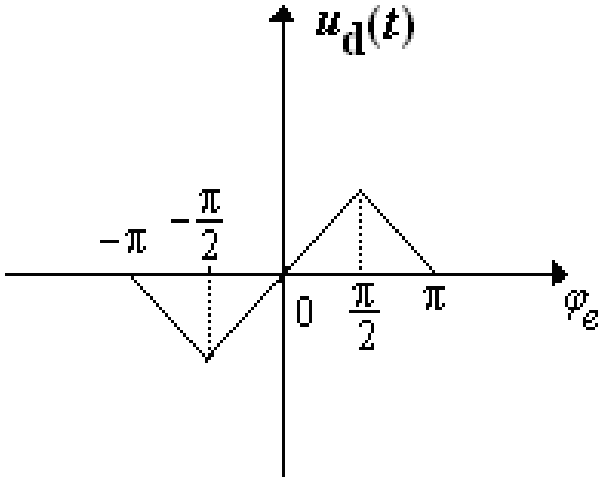
$$A_p = \frac{K_p}{2} U_{rm} U_{0m}$$

$$u_d(t) = A_p \sin(\varphi_e(t))$$



当相位差$\varphi_e(t)$较小时，有

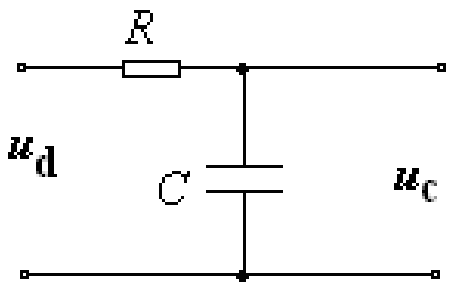
$$u_d(t) = A_p \ \varphi_e(t)$$



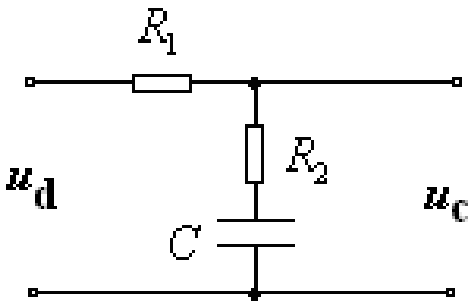
2. 低通滤波器

常用的低通滤波器有以下三种形式：

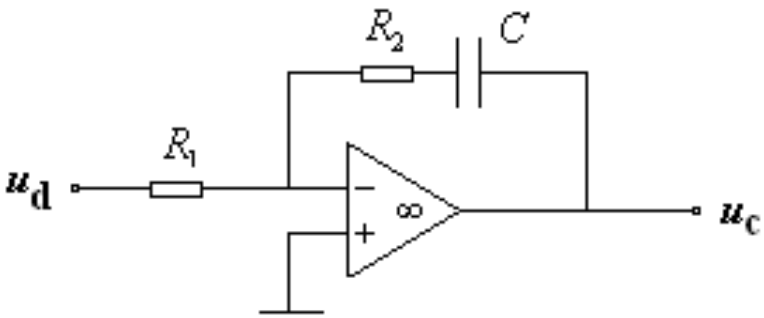
- ★ *RC*积分滤波器
- ★ *RC*比例积分滤波器
- ★ 有源比例积分滤波器



$$F(s)=\frac{U_c(s)}{U_d(s)}=\frac{1}{1+s\tau}$$



$$F(s)=\frac{U_c(s)}{U_d(s)}=\frac{1+s\tau_2}{1+s(\tau_1+\tau_2)}$$



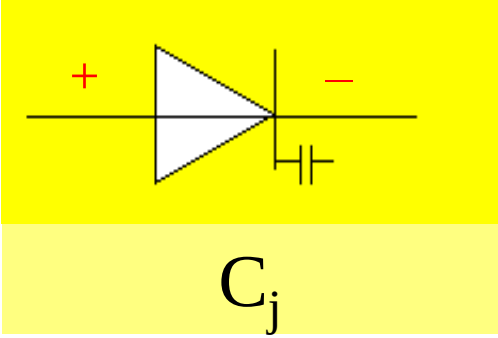
$$F(s)=-\frac{R_2+\frac{1}{sC}}{R_1}=\frac{1+s\tau_2}{s\tau_1}$$

3. 压控振荡器(VCO)

变容二极管

它的结电容受反向外加电压控制而变化。

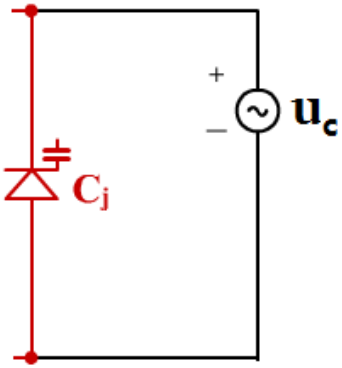
$$C_j = \frac{C_{j0}}{(1-\frac{u_c}{U_D})^{\gamma}}$$



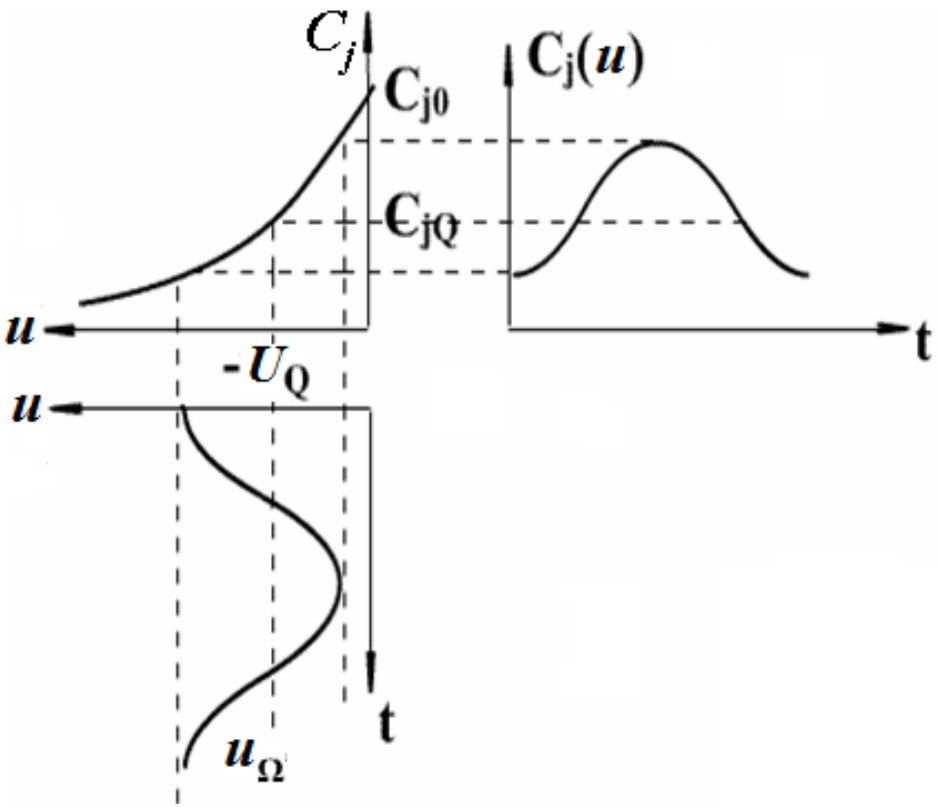
C_{j0} —— 反向电压 u= 0 时的结电容

U_D —— PN结内建电位差(很小)

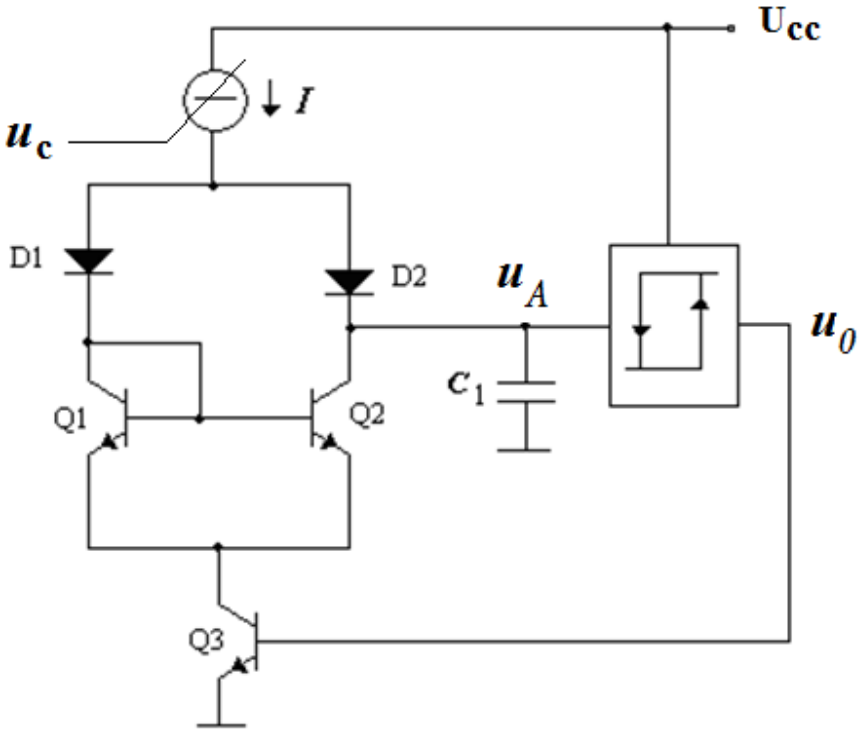
γ —— 电容的变化指数

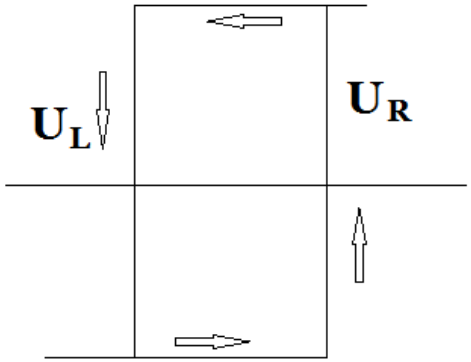


$$C_j = \frac{C_{j0}}{(1-\frac{u_c}{U_D})^{\gamma}}$$



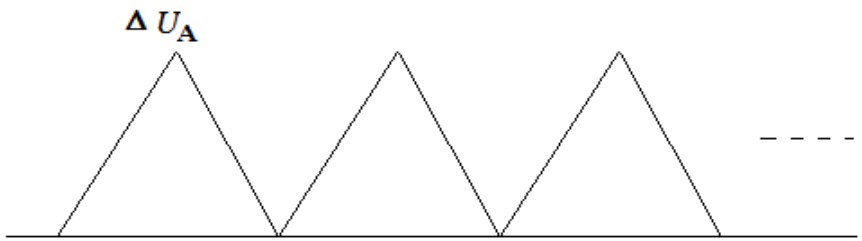
集成压控振荡器





可控恒流源电压u_c的控制电容C_1的充放电速度，改变压控振荡器的振荡频率。

集成压控振荡器的原理图



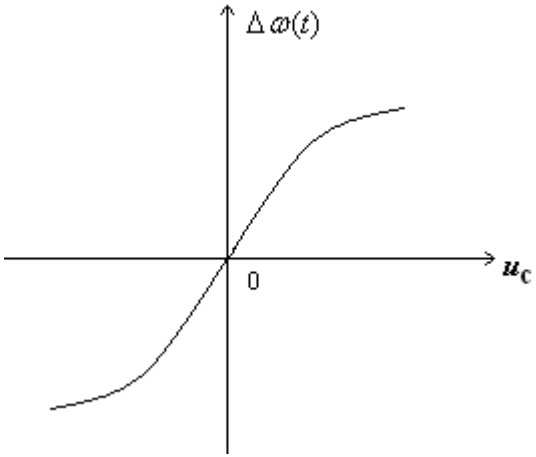
$$I = C_1 \frac{\Delta U_A}{\Delta t}$$

$$f = \frac{1}{2\Delta t} = \frac{I}{2\Delta U_A C_1} = \frac{I}{2(U_R - U_L)C_1}$$

$$I = g_m u_c \qquad f = \frac{g_m u_c}{2(U_R - U_L)C_1} = A_c u_c$$

VCO的角频率变化可近似由线性方程表示

$$\Delta\omega(t)=A_c\, u_c(t)$$



VCO的瞬时频率：

$$\omega_0(t)=\omega_{c0}+\Delta\omega(t)=\omega_{c0}+A_c\, u_c(t)$$

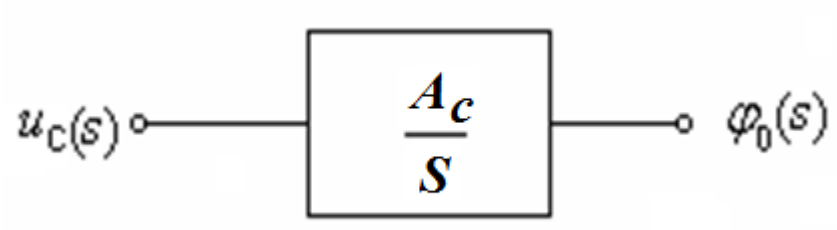
VCO的瞬时相位

$$\varphi(t)=\int\omega_0(t)dt=\omega_{c0}t+\int A_c u_c(t)dt$$

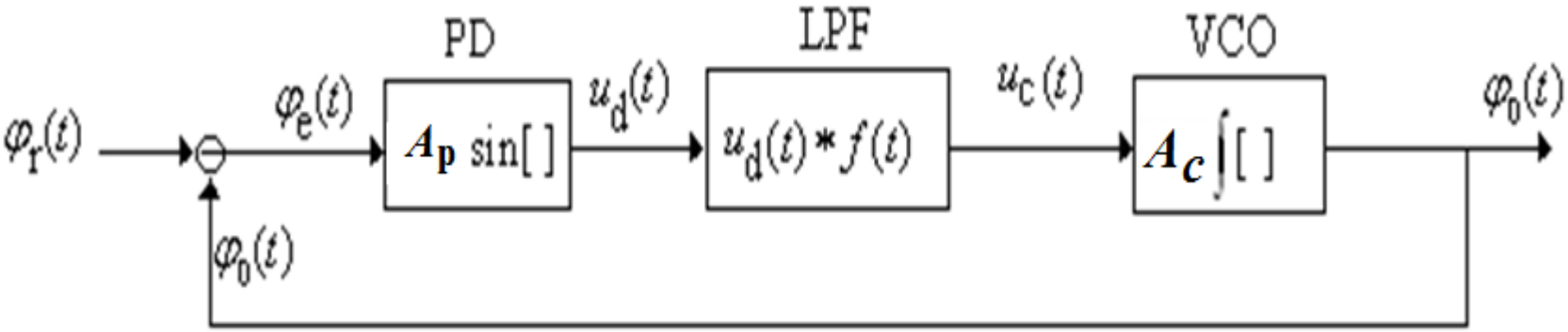
$$\varphi_0(t)=\int A_c u_c(t)dt$$

就$\varphi_0(t)$和$u_c(t)$之间的关系而言，VCO是一个理想积分器。

$$\varphi_0(s)=A_c\frac{u_c(s)}{s}$$

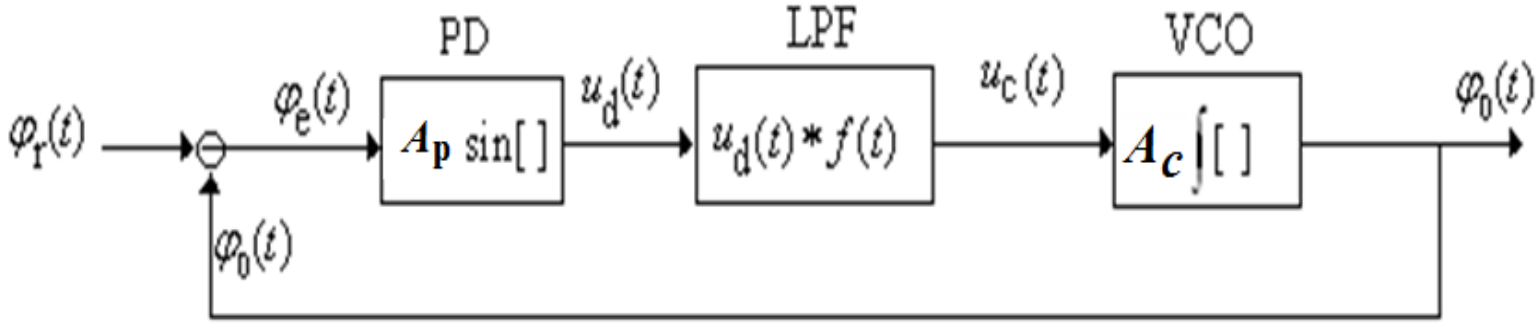


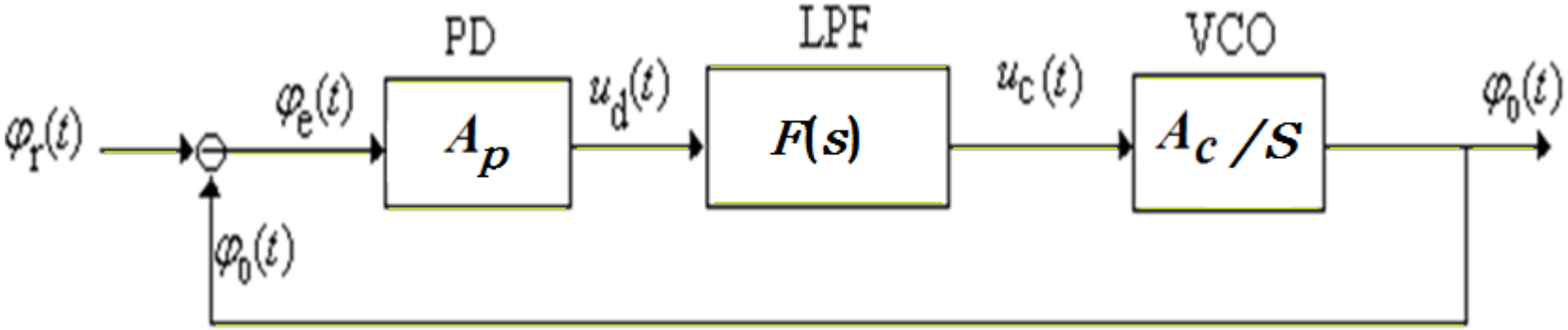
4. 锁相环的线性化分析

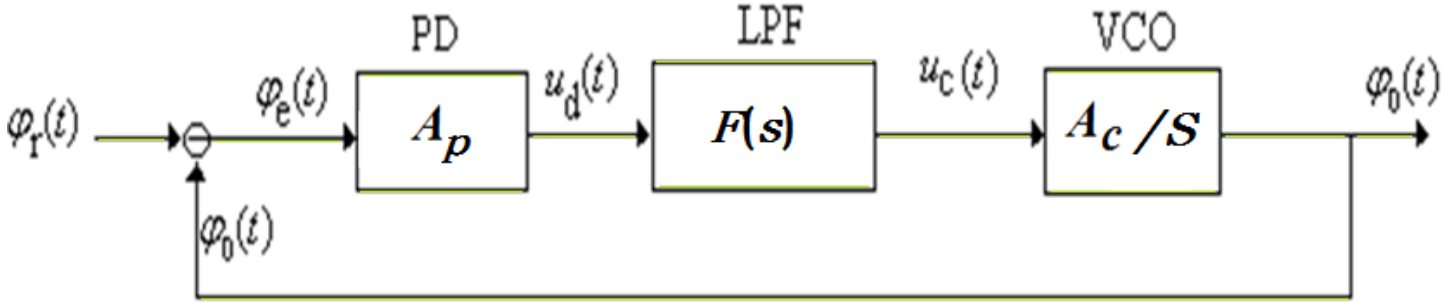


PLL的相位模型

$$\varphi_0(t) = A_p \sin[\varphi_r(t) - \varphi_0(t)] \times F(s) \times A_c \times \frac{1}{s}$$


PD
LPF
VCO
$\varphi_r(t)$
$\varphi_e(t)$
$A_p \sin[\,]$
$u_d(t)$
$u_d(t) * f(t)$
$u_c(t)$
$A_c \int[\,]$
$\varphi_0(t)$
$\varphi_0(t)$


PD
LPF
VCO
$\varphi_r(t)$
$\varphi_e(t)$
A_p
$u_d(t)$
$F(s)$
$u_c(t)$
A_c / S
$\varphi_0(t)$
$\varphi_0(t)$



$$u_d(t) = A_p\ \varphi_e(t) = A_p\ [\varphi_r(t) - \varphi_0(t)]$$

$$\Delta\omega = A_c\, u_c(t)$$

$$\omega_0 = \omega_{c0} + \Delta\omega = \omega_{c0} + A_c\, u_c(t)$$

$$\varphi_0(t) = \int \omega_0(t)dt = \omega_{c0}t + \int A_c u_c(t)dt$$

$$\varphi_e = \varphi_r - \varphi_0 = \varphi_r - A_p F(s)\frac{A_c}{s}\varphi_e$$

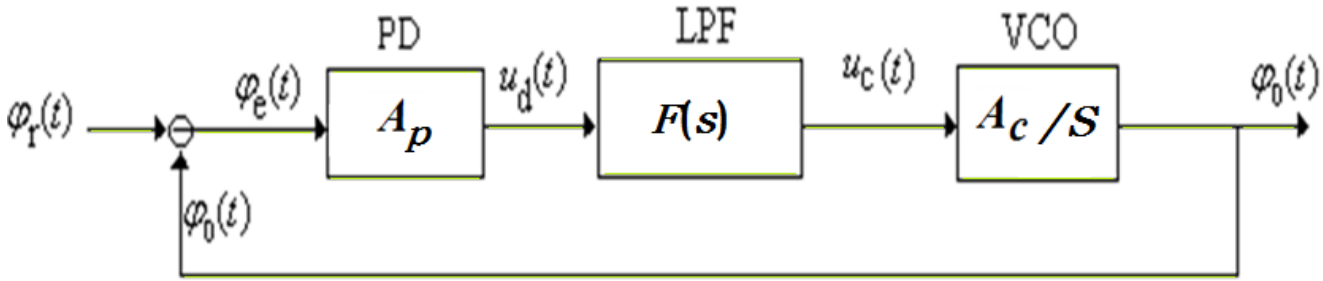
$$s\varphi_e + A_p F(s) A_c \varphi_e = s\varphi_r$$

$$\varphi_e = \frac{s}{s + A_p A_c F(s)}\varphi_r$$

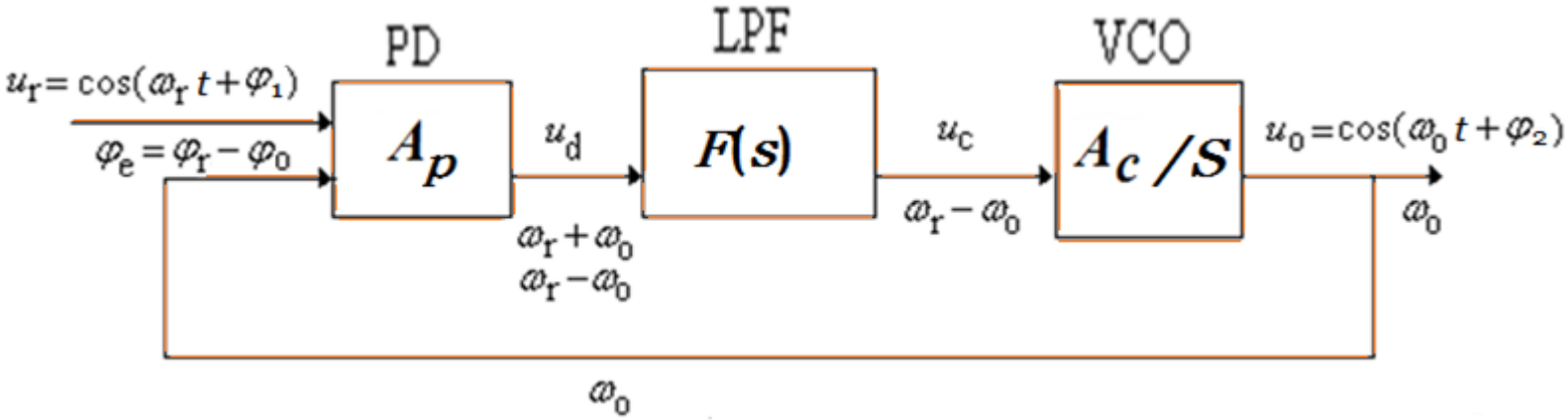
$$\varphi_0 = \frac{A_p A_c F(s)}{s}\varphi_e$$

闭环传递函数

$$\frac{\varphi_0}{\varphi_r} = \frac{A_p A_c F(s)}{s + A_p A_c F(s)} = H(s)$$



PLL环路工作原理



最初这两个频率不相等 $\omega_0 \neq \omega_r$

VCO的频率偏移量Δω与差频信号的极性和振幅成比例

VCO初始频率为ω_{c0} **VCO的输出频率$\omega_0 = \omega_{c0} + \Delta\omega$**

经过几个周期后，VCO的输出频率和外部输入参考频率ω_r相等，即$\omega_o = \omega_r$

锁相环的重要性质

- ◆环路锁定后，没有剩余频差。即$\omega_0=\omega_r$
- ◆跟踪特性。环路锁定后，当参考频率ω_r稍有变化时，VCO的频率立即发生相应的变化，最终$\omega_0=\omega_r$
- ◆滤波特性。锁相环通过环路滤波器的作用，具有窄带滤波特性。这种窄带滤波特性是任何LC、RC、石英晶体、陶瓷片等滤波器难以达到的。
- ◆ 易于集成化。

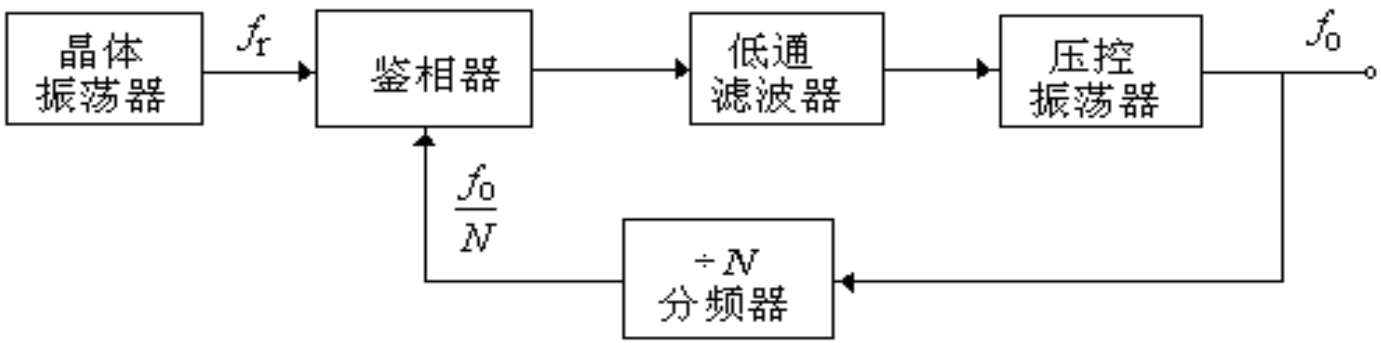
§4.3 锁相频率合成器

利用频率跟踪特性，锁相电路可实现分频、倍频、混频等频谱变换功能。

综合这几种功能又可构成频率合成器与标准频率源。

结构简单、成本低。

(1)单环锁相频率合成器



环中÷N分频器采用可编程的程序分频器

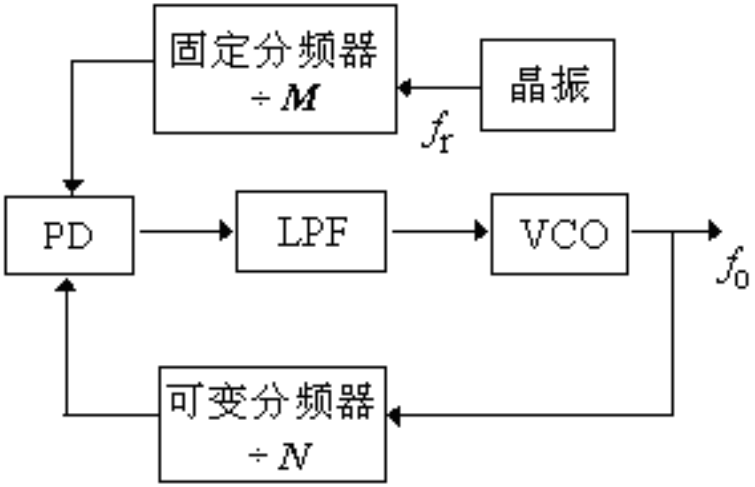
输出频率：

$$f_0 = N f_r$$

例2：

若题图所示频率合成器框图中的晶体振荡器频率为5MHz，固定分频器的分频比M=50，可变分频器的分频比N=1~100。

求输出信号频率的变化范围和最小频率间隔。



解 晶振输出频率f_r经50分频作为锁相环的输入信号

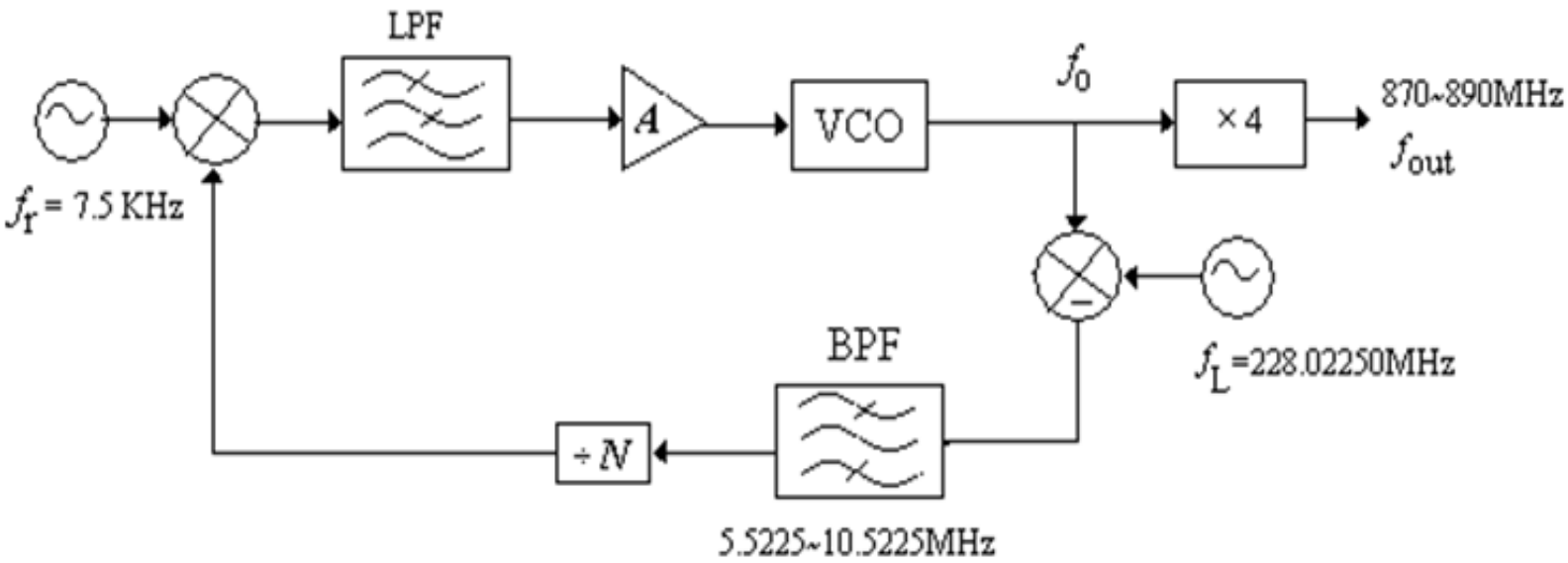
$$f_i = \frac{f_r}{M} = \frac{5\times10^6}{50} = 100\text{kHz}$$

当N=1~100变化时，输出信号频率的变化范围

$$f_0 = (1\sim100)\times100\text{kHz} = 100\text{kHz}\sim10\text{MHz}$$

最小频率间隔： $f_{ch} = f_i = 100\text{kHz}$

(2) 带混频器的频率合成器



工作频率f_{out}=870 ~ 890MHz，　　求分频数N=?

VCO的频率范围应是(870 ~ 890)/4=(217.5 ~ 222.5)MHz

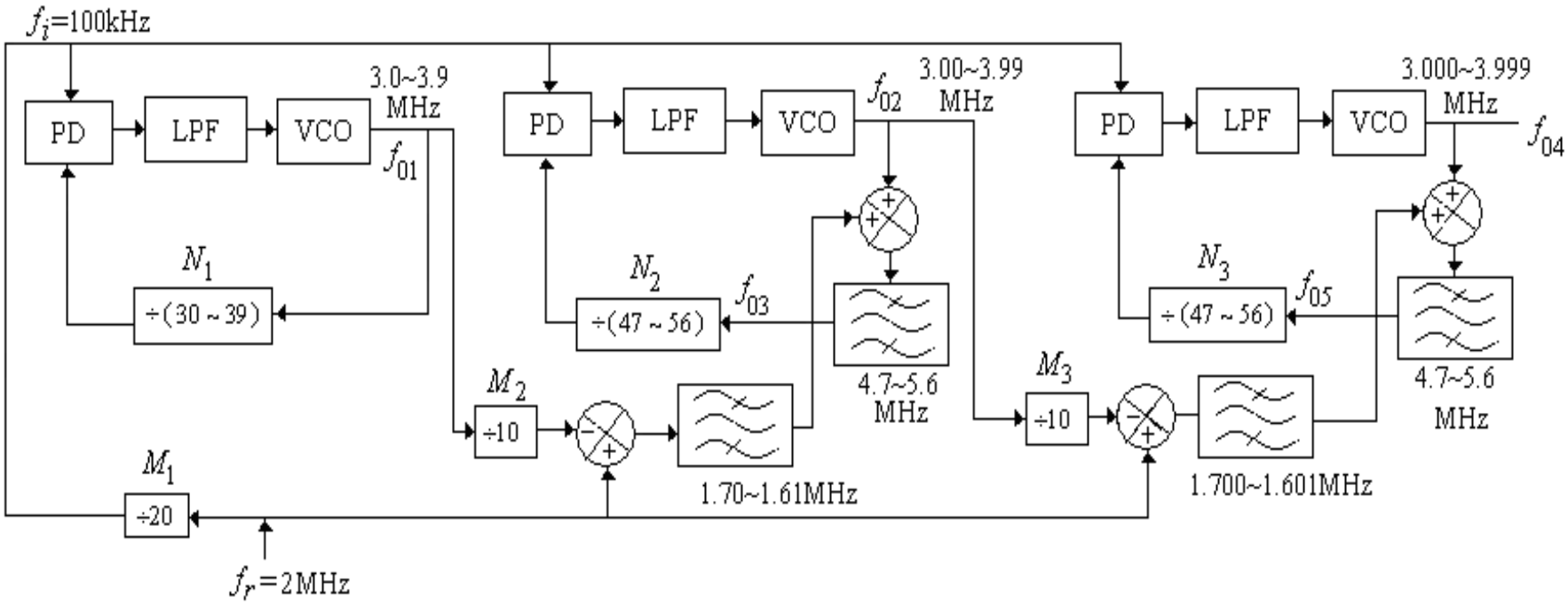
下混频，得到f_L - f_0 = (5.5225 ~ 10.5225)MHz

N = (5.5225 ~ 10.5225)/0.0075 = 736 ~ 1403

分频数N=736 ~ 1403

(3) 多环频率合成器

晶振产生的参考频率为2MHz，要求输出频率范围为3.000～3.999MHz，频率间隔为1kHz

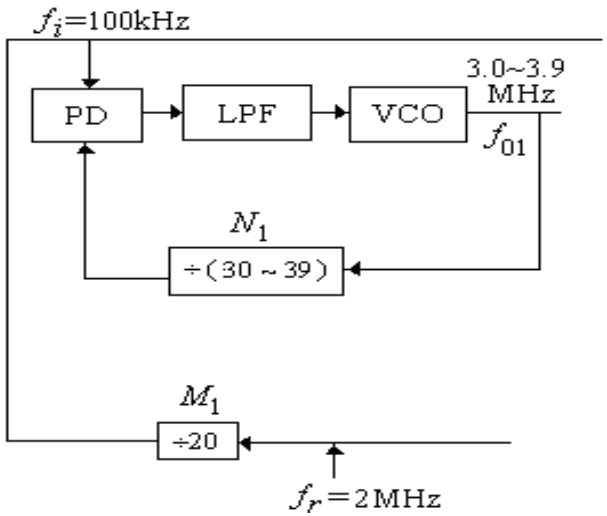


第1个环输出　　2MHz的参考频率首先经20分频，得到f_i=100kHz

$$f_{01} = N_1 f_i = \frac{N_1}{M_1} f_r = \frac{30 \sim 39}{20} \times 2\text{MHz} = (3.0 \sim 3.9)\text{MHz}$$

频率间隔

$$f_{ch1} = \frac{f_r}{M_1} = f_i = 100\text{kHz}$$



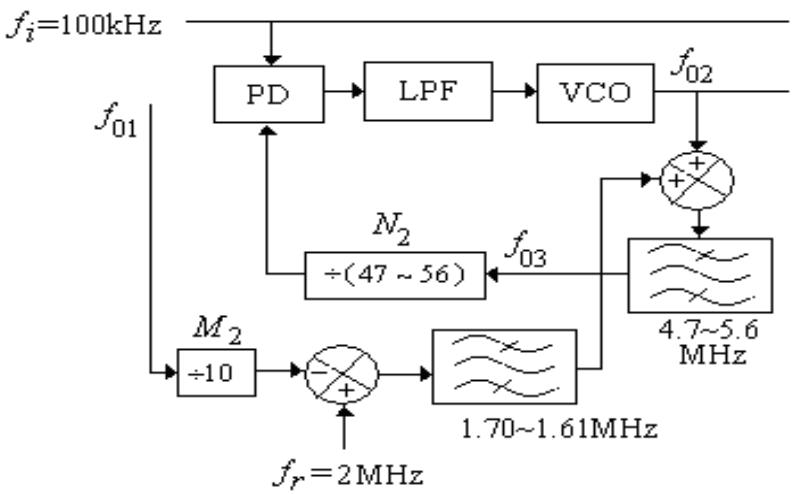
第2个环 设输出为f_{02}，中间量f_{03}

$$f_{03} = N_2 f_i = (47\sim56)\times100\text{kHz} = (4.7\sim5.6)\text{MHz}$$

又 $f_{03} = f_{02} + (f_r - f_{01}/M_2)$

所以有 $f_{02} = f_{03} - f_r + f_{01}/10 = (3.00\sim3.99)\text{MHz}$

频率间隔$f_{ch2} = f_{ch1}/10 = 10\text{kHz}$



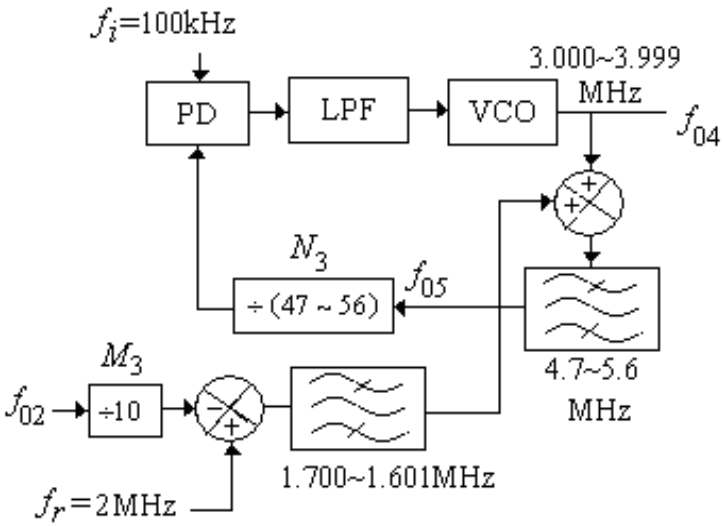
第3个环　设输出为f04，中间变量f05，有

$f_{05} = N_3 f_i = (47\sim56)\times100\text{kHz} = (4.7\sim5.6)\text{MHz}$

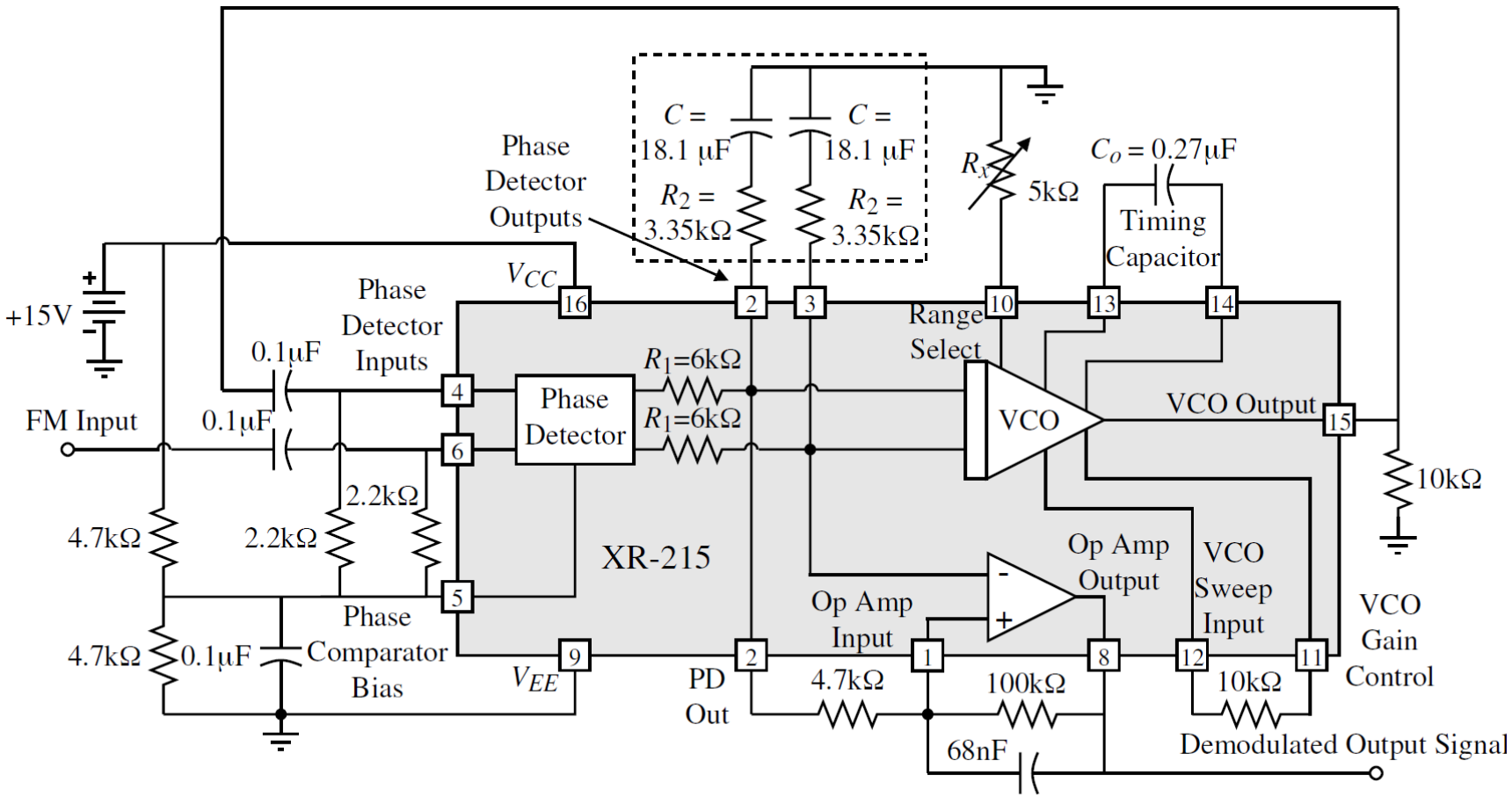
又$f_{05} = f_{04} + (f_r - f_{02}/M_3)$

$f_{04} = f_{05} - f_r + f_{02}/10 = (3.000\sim3.999)\text{MHz}$

频率间隔$f_{ch3} = f_{ch2}/10 = 1\text{kHz}$

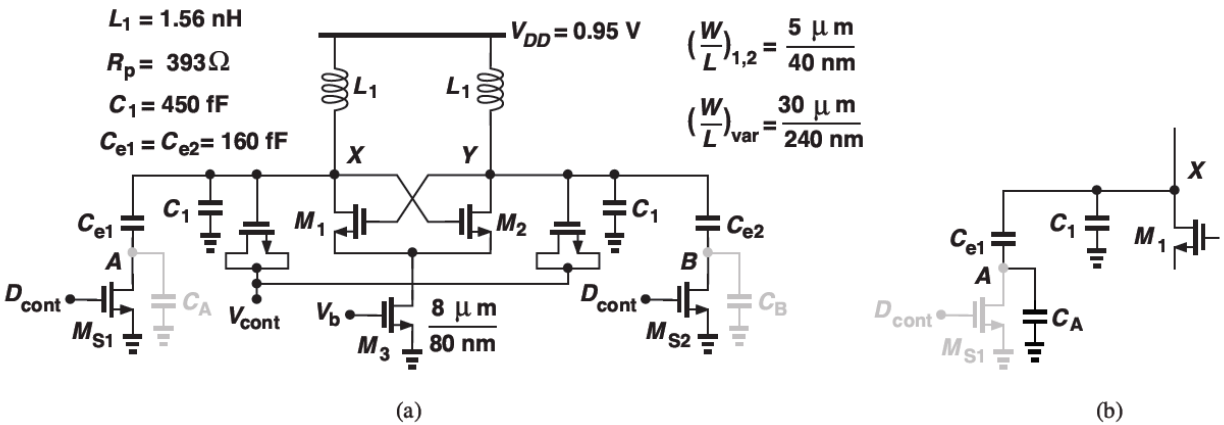


§4.3 PLL频率综合器电路的案例、设计与应用

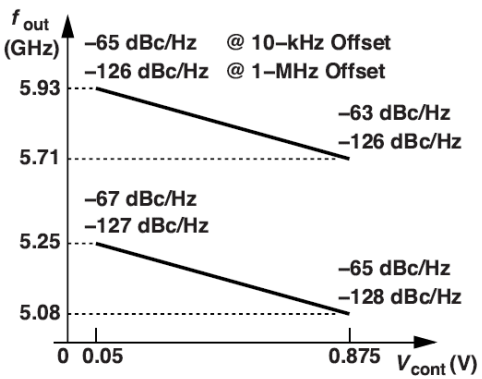


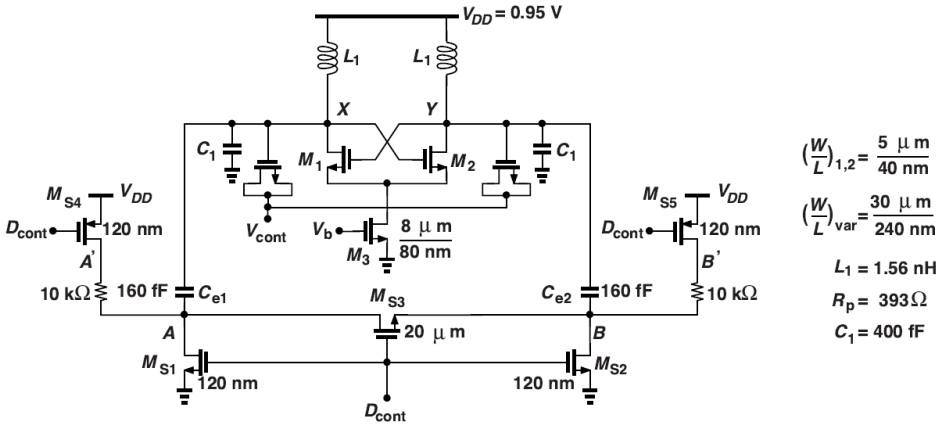
§4.3 PLL频率综合器电路的案例、设计与应用

- LC压控振荡器



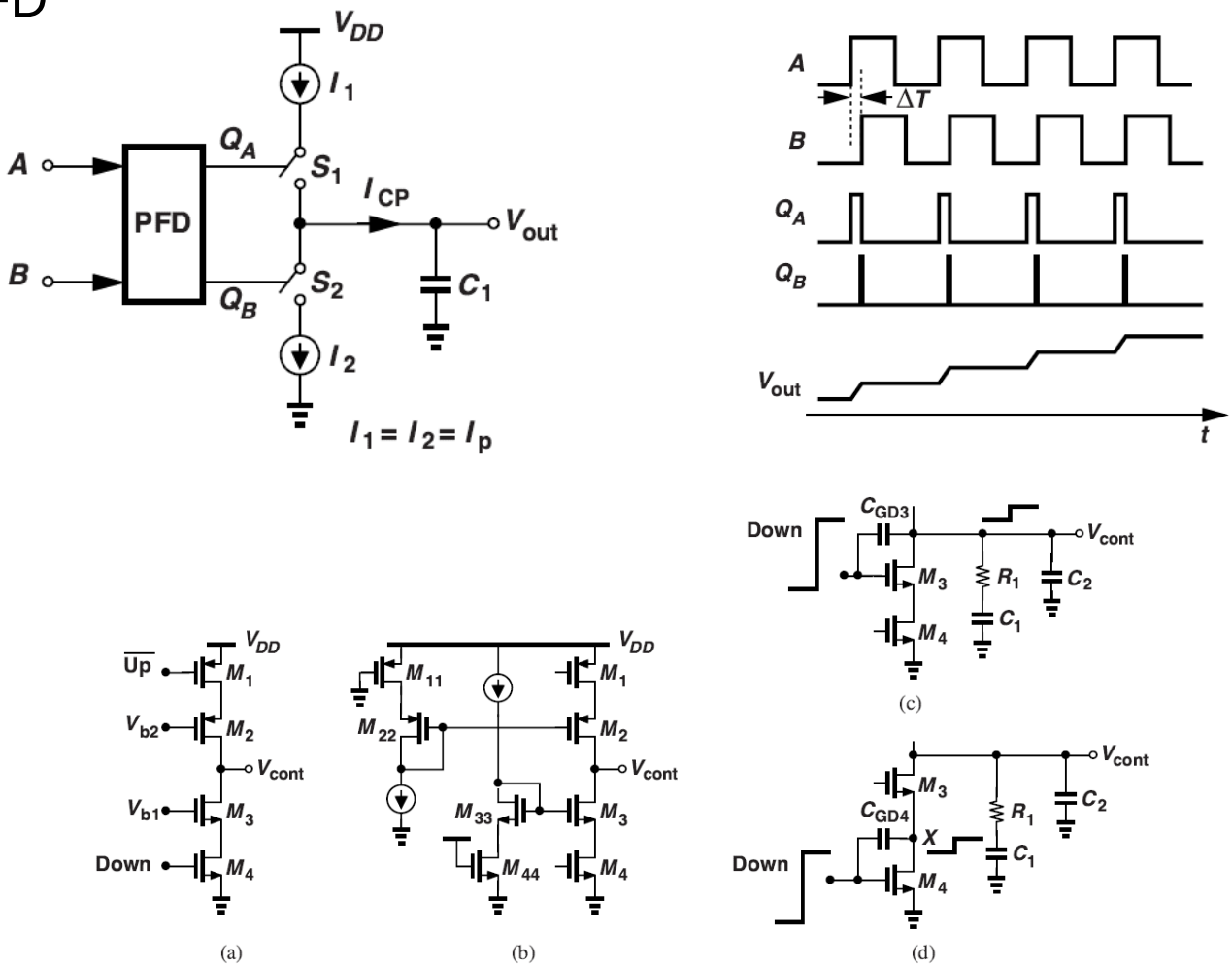
(a) VCO of Fig. 5.45(a) with discrete tuning added, and (b) effect of switch parasitic capacitance.





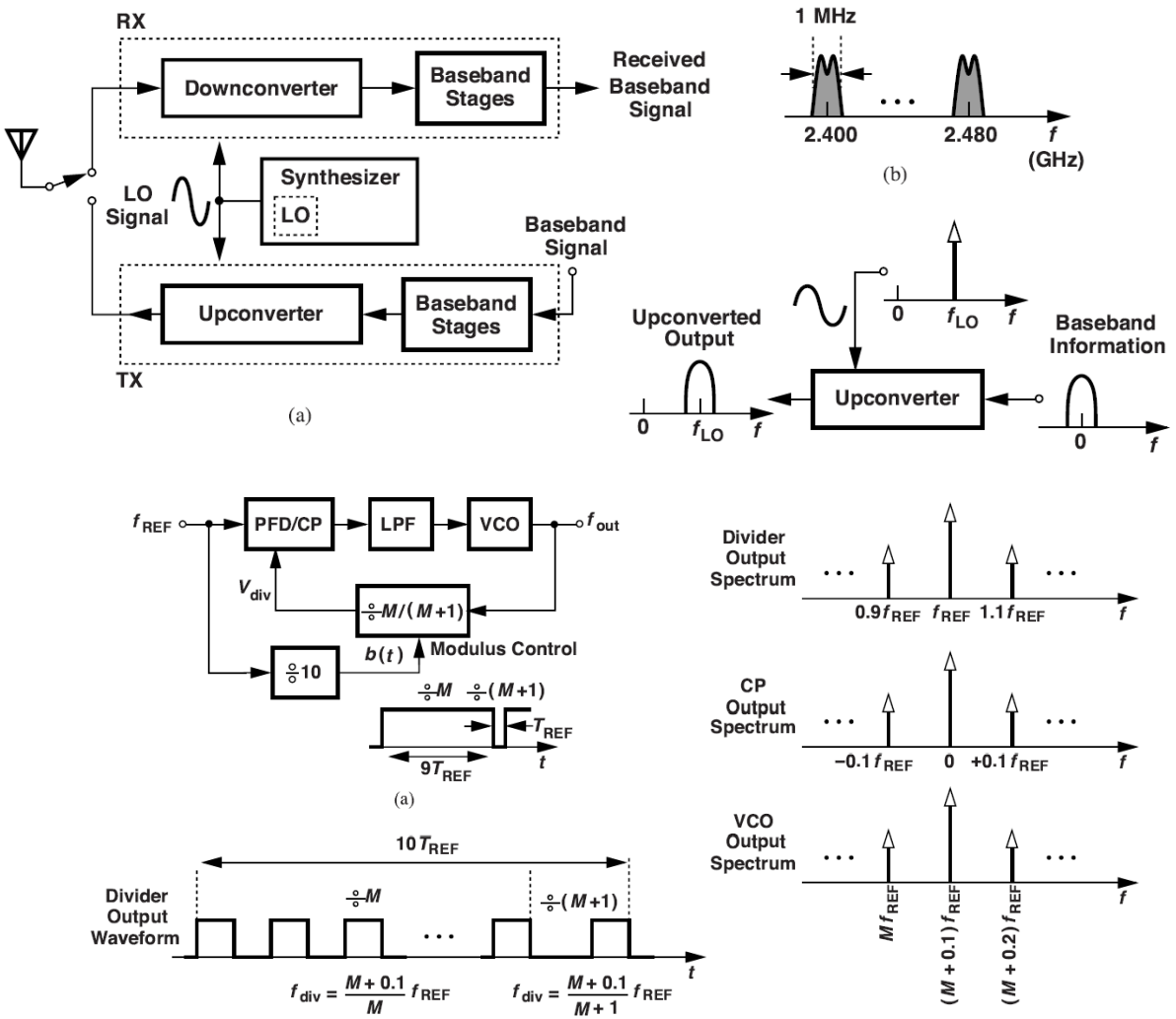
§4.3 PLL频率综合器电路的案例、设计与应用

- CP-PFD

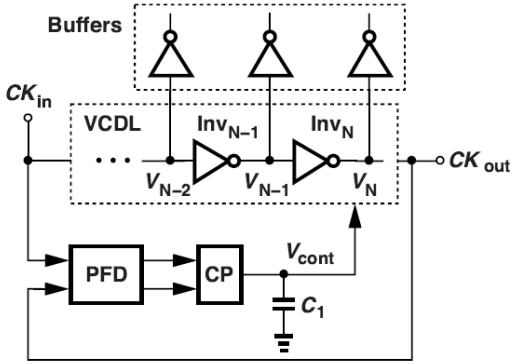


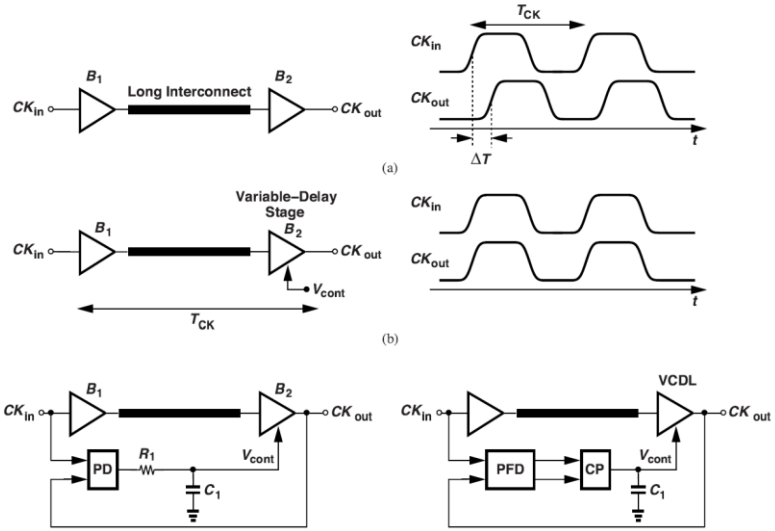
(a) Source-switched CP, (b) bias implementation, (c) clock feedthrough in drain-switched CP, and (d) clock feedthrough in source-switched CP.

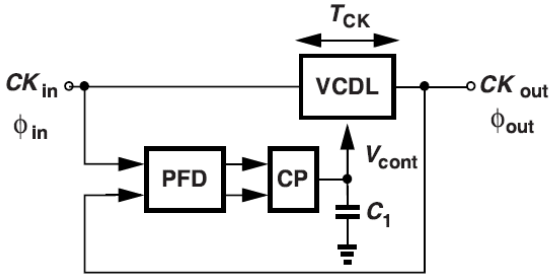
高纯度射频信号综合生成



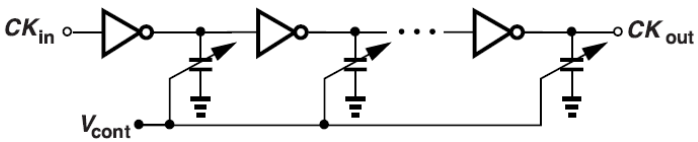
延迟锁定与时钟去偏斜



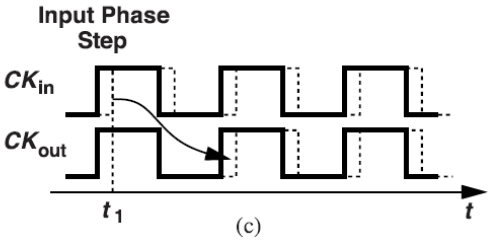




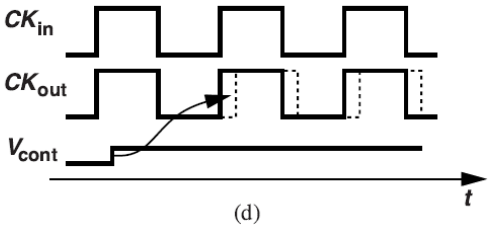
(a)



(b)



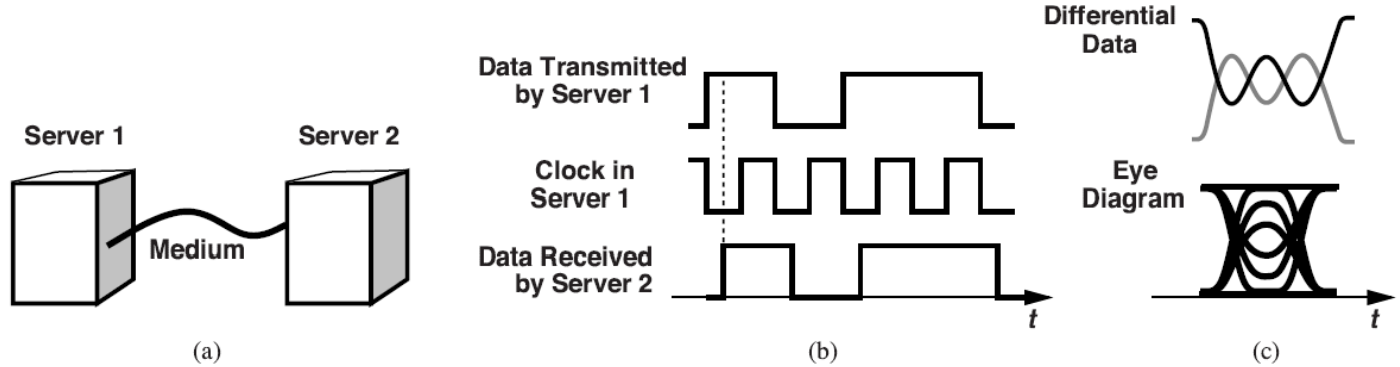
(c)

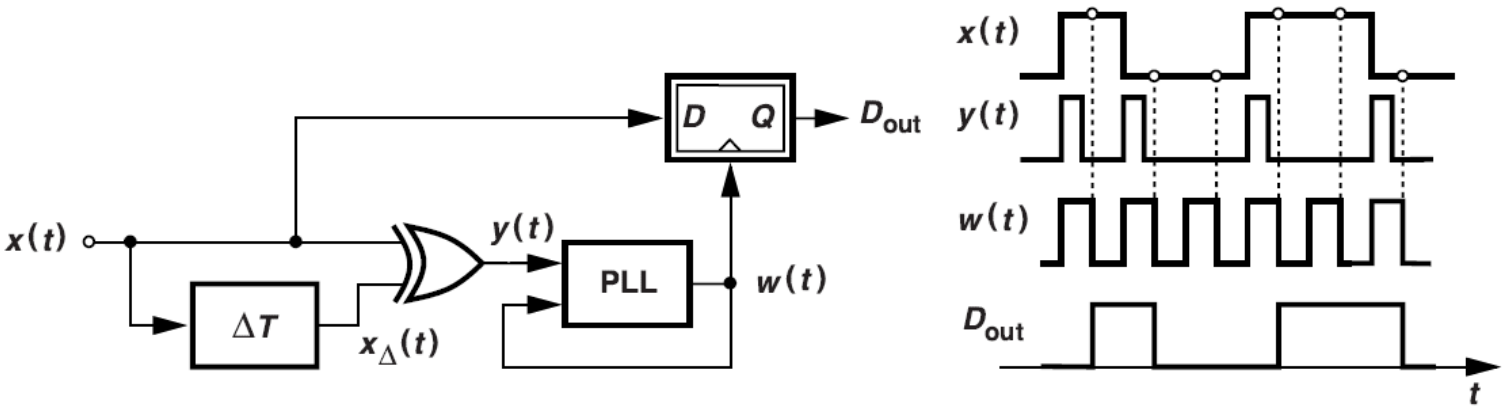


(d)

§4.3 PLL频率综合器电路的案例、设计与应用

时钟与数据恢复电路





调频信号的调制与解调

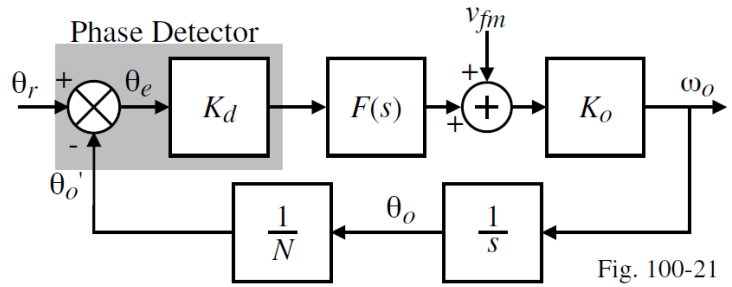


Fig. 100-21

$$\frac{\omega_o(s)}{V_{fm}(s)} = \frac{sK_o}{s + \frac{K_v F(s)}{N}} \Rightarrow \frac{\omega_o(s)}{V_{fm}(s)} = \frac{sK_o}{s + \frac{K_v}{N}}$$

Highpass filter with a gain of K_o and $\omega_{\text{-3dB}} = \frac{K_v}{N}$

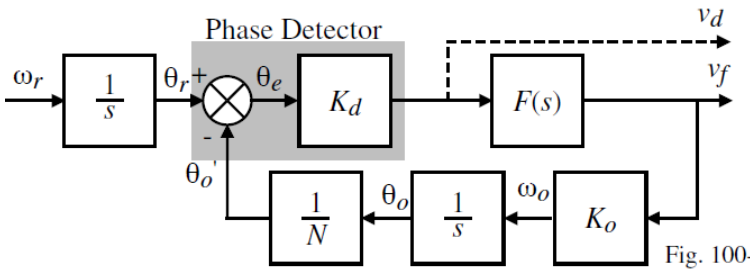


Fig. 100-23

$$\frac{V_f(s)}{\omega_r(s)} = \frac{K_d F(s)}{s + \frac{K_v F(s)}{N}} \Rightarrow \frac{V_f(s)}{\omega_r(s)} = \frac{K_d}{s + \frac{K_v}{N}}$$

Lowpass filter with a gain of $\frac{N}{K_o}$ and $\omega_{\text{-3dB}} = \frac{K_v}{N}$

调相信号的调制与解调

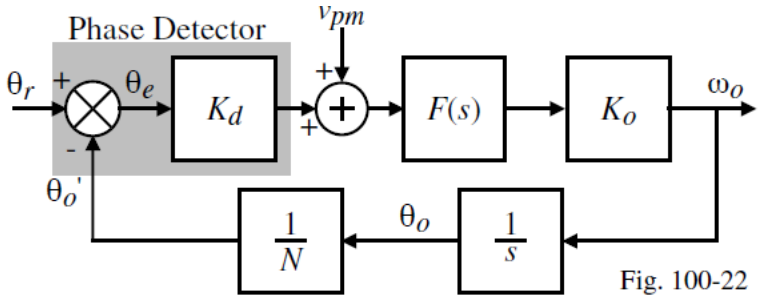
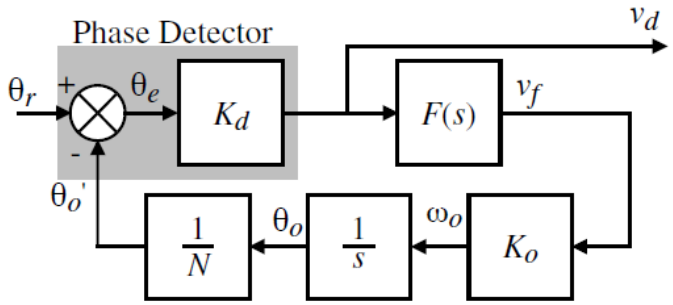


Fig. 100-22

$$\frac{\theta_o(s)}{V_{pm}(s)} = \frac{F(s)K_o}{s + \frac{K_v F(s)}{N}} \Rightarrow \frac{\theta_o(s)}{V_{pm}(s)} = \frac{K_o}{s + \frac{K_v}{N}}$$

Lowpass filter with a gain of $\frac{N}{K_d}$ and $\omega_{\text{-3dB}} = \frac{K_v}{N}$



$$\frac{V_d(s)}{\theta_r(s)} = \frac{sK_d}{s + \frac{K_v F(s)}{N}} \Rightarrow \frac{V_d(s)}{\theta_r(s)} = \frac{sK_d}{s + \frac{K_v}{N}}$$

Highpass filter with a gain of K_d and $\omega_{\text{-3dB}} = \frac{K_v}{N}$